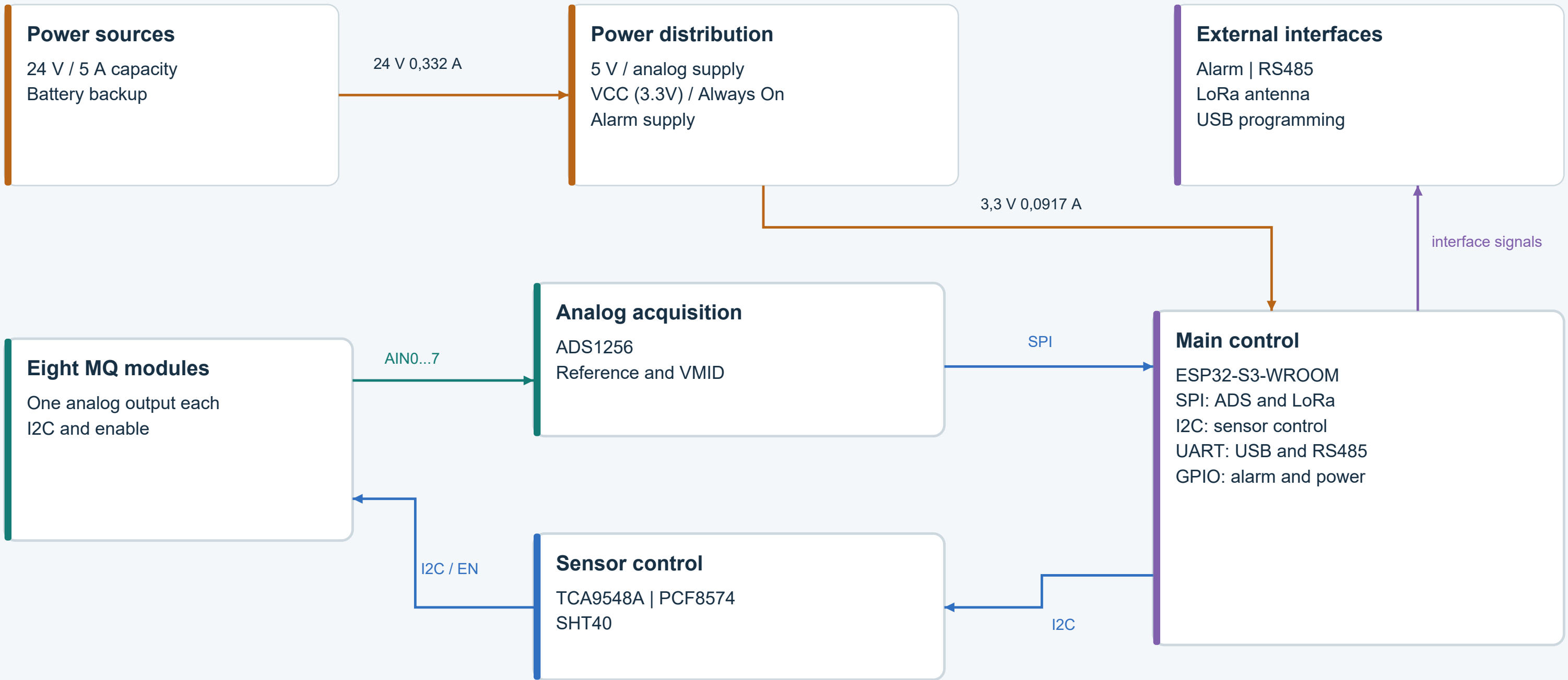


01 System architecture

24 V supply case: all eight sensors active, ESP32 RF off, LoRa transmitting, alarm off.

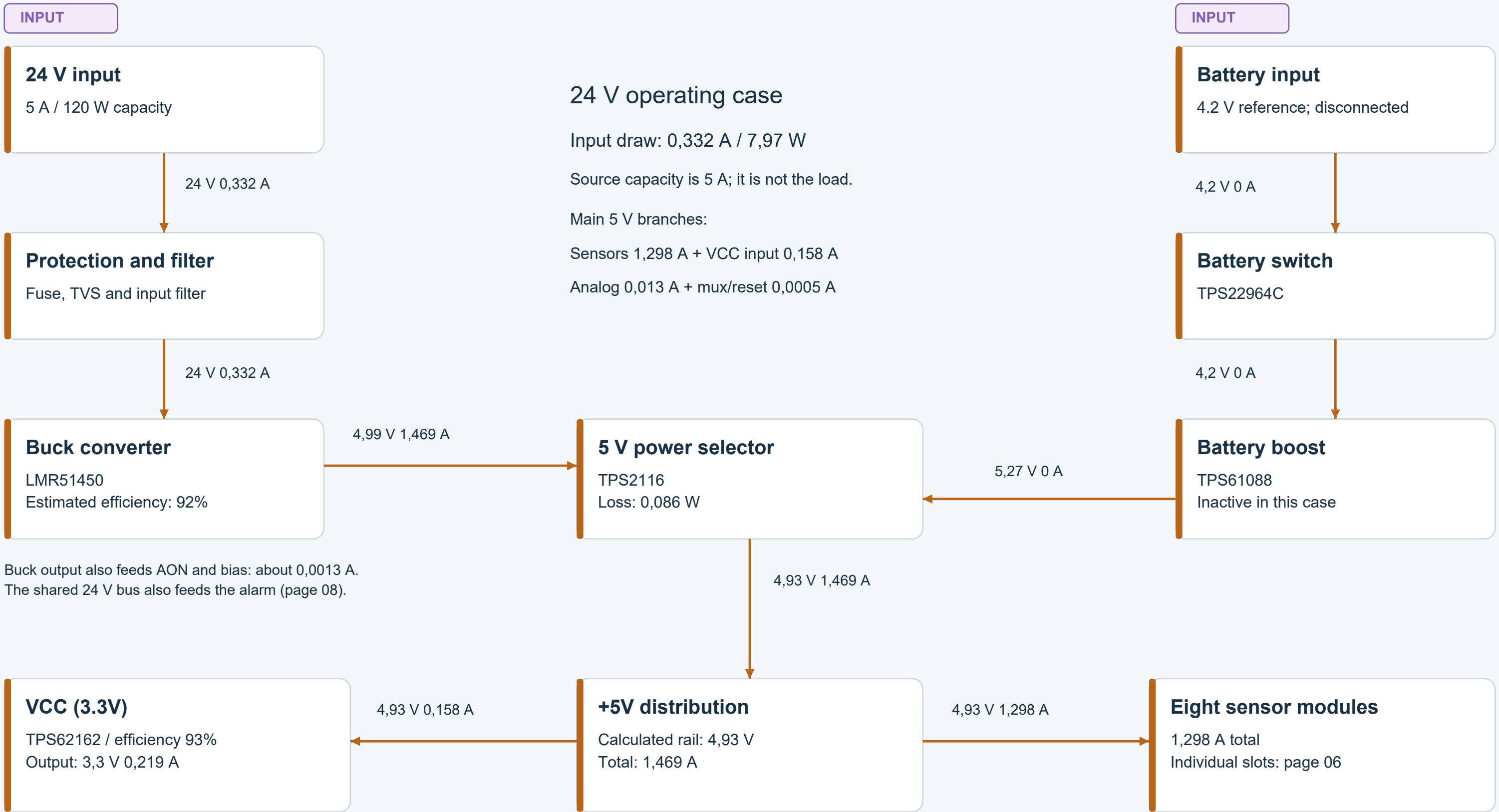


Calculated reference case - not a measured or maximum load

All brown-line values are estimates for this case. Heater resistance, IC references and passive-load allowances are included.
Converter curves: about 92% for 24 V to 5 V and 93% for 5 V to 3.3 V. Source allocation and exclusions: page 09.

02 Main power distribution

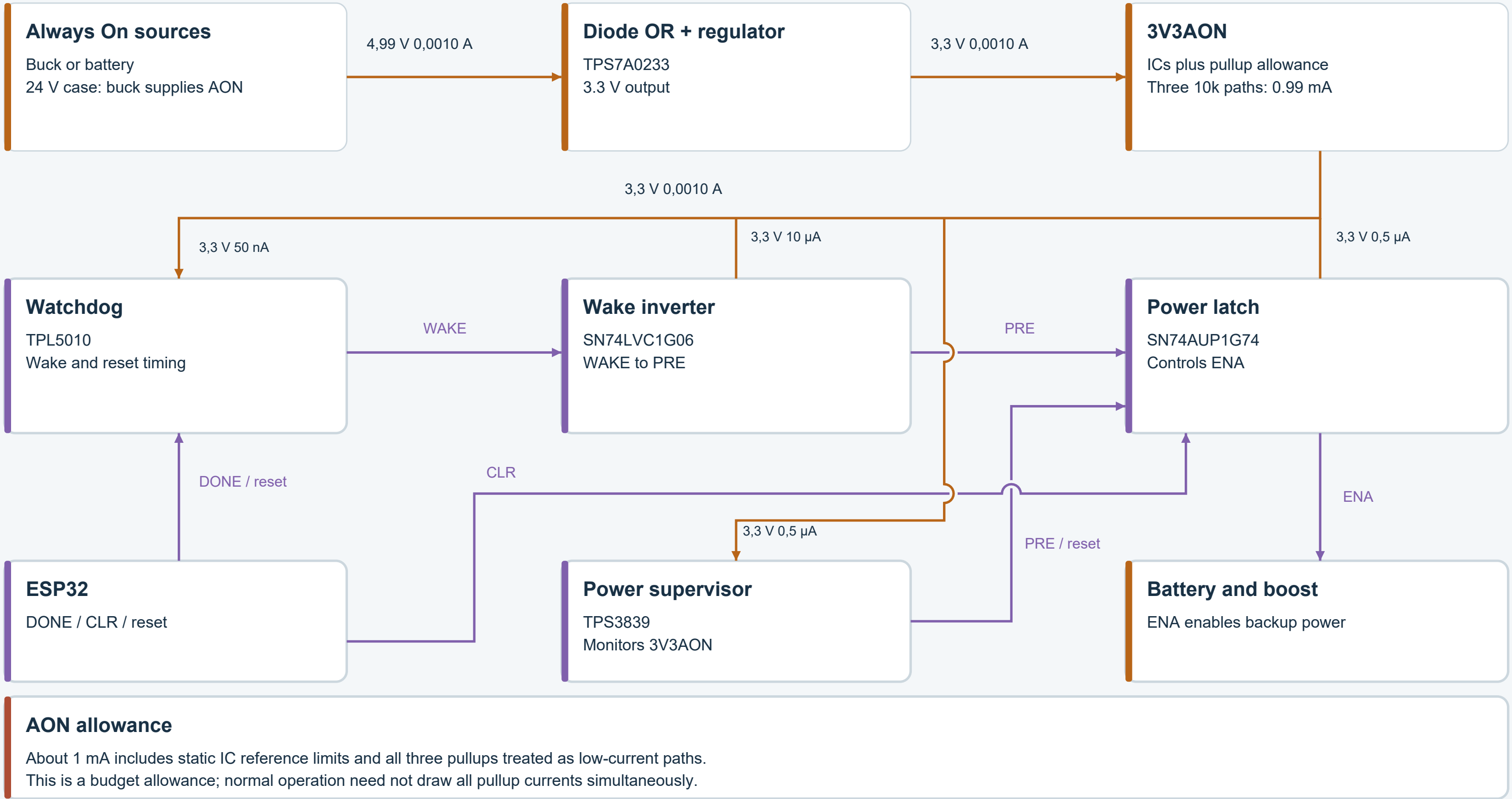
Main source at left; battery at right. Values apply to the 24 V case; battery contribution is zero.



Nominal resistor values and typical switch resistance set the displayed voltage drops. Wiring/filter losses and startup are excluded.

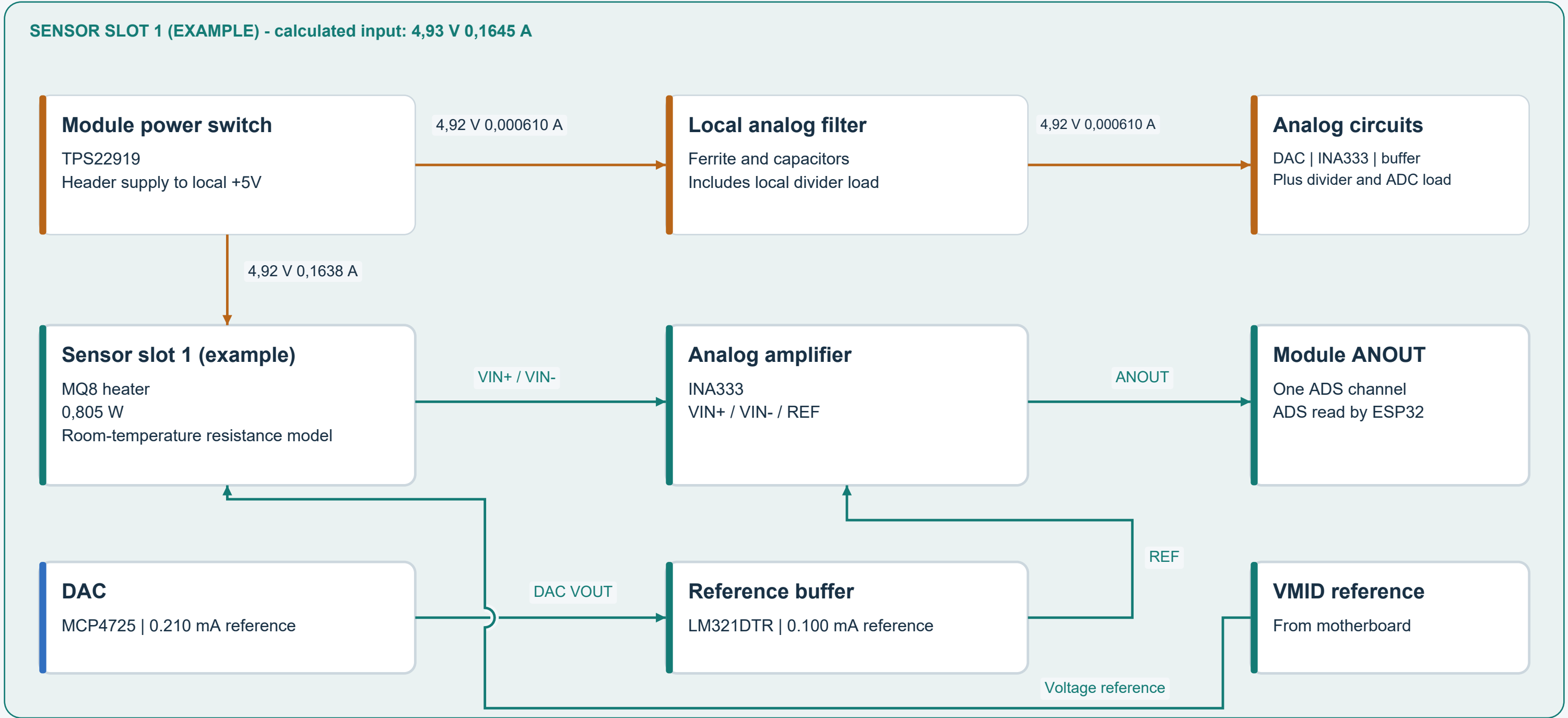
03 Always On, watchdog, and latch

Purpose: keep supervision and wake control available when the main switched circuitry is off.



04 Sensor module: functional path

One example of eight modules. Local analog supply is separate from the motherboard analog supply.

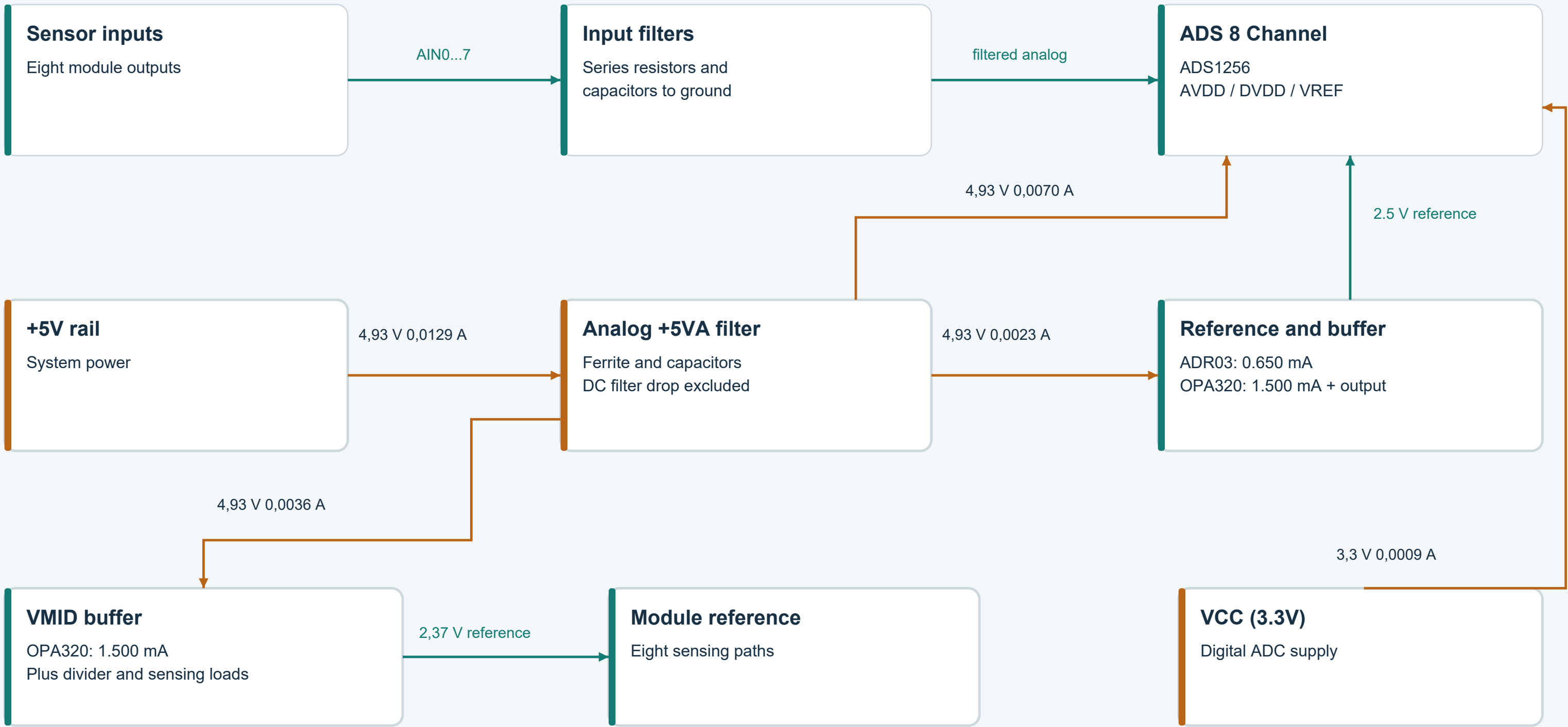


Module allocation

INA333: 0.050 mA; local divider: about 0.246 mA; one ADC-input allowance is shared across eight modules.
Header current also includes switch bias. Sensor sensing current is supplied by motherboard VMID (page 05).

05 Analog acquisition and voltage reference

Power labels include reference IC currents and calculated passive-load allowances.

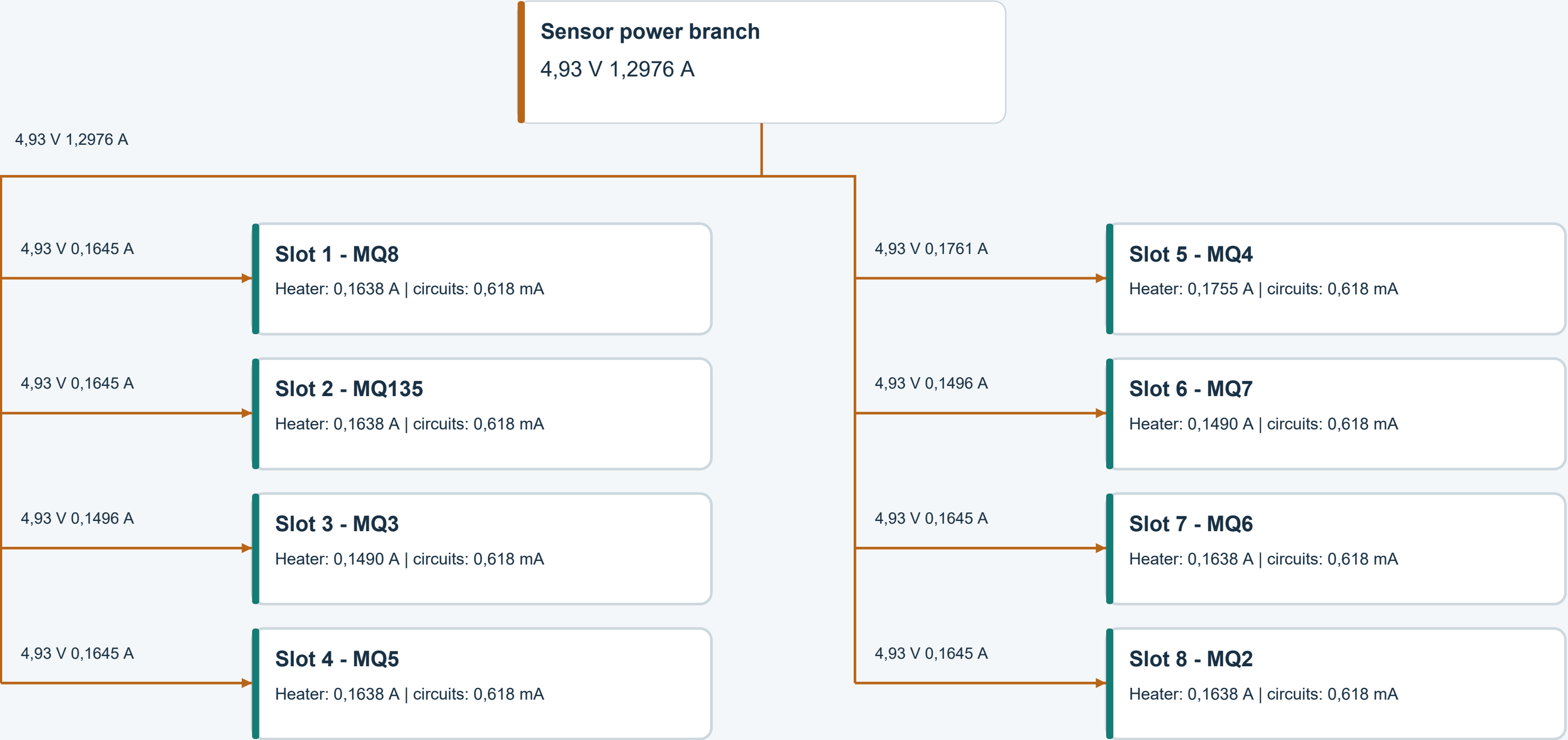


VMID sensing allowance: $1,896\text{ mA}$ for all eight modules; it is not a sensor input signal.

ADS current uses the 7.68 MHz datasheet benchmark for the 8 MHz schematic as an estimate; PGA 1, input buffer off.

06 Eight sensor modules: load split

Current at each module header includes its heater, analog electronics and switch bias.

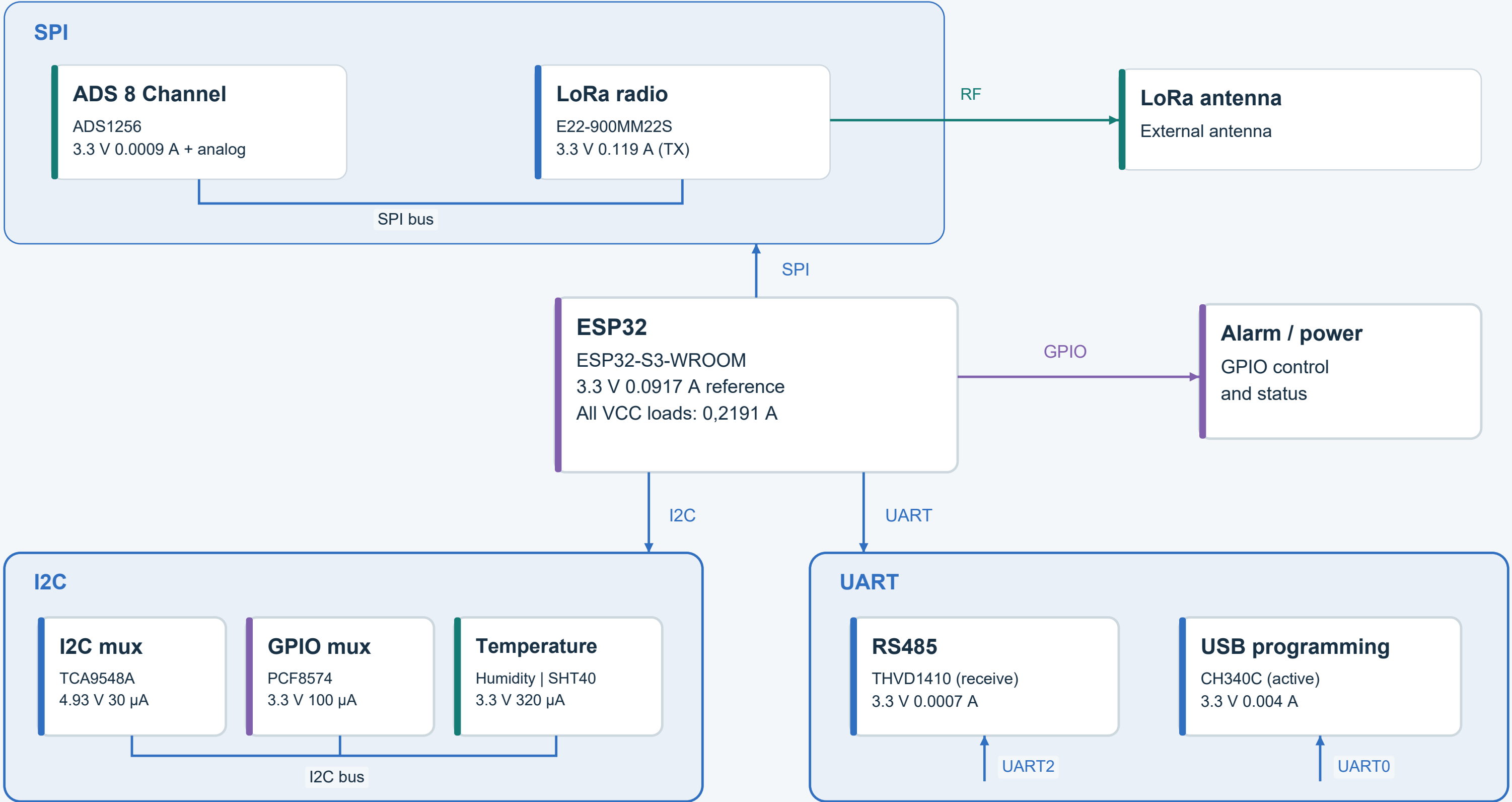


Heater model and module control

Heaters use datasheet room-temperature resistance; calculated local voltage is about 4.92 V after switches.
I2C selects each module DAC; PCF8574 enables module power. MQ7 here uses the supplied fixed-supply circuit, not a low/high cycle.

07 ESP32 communications and connections

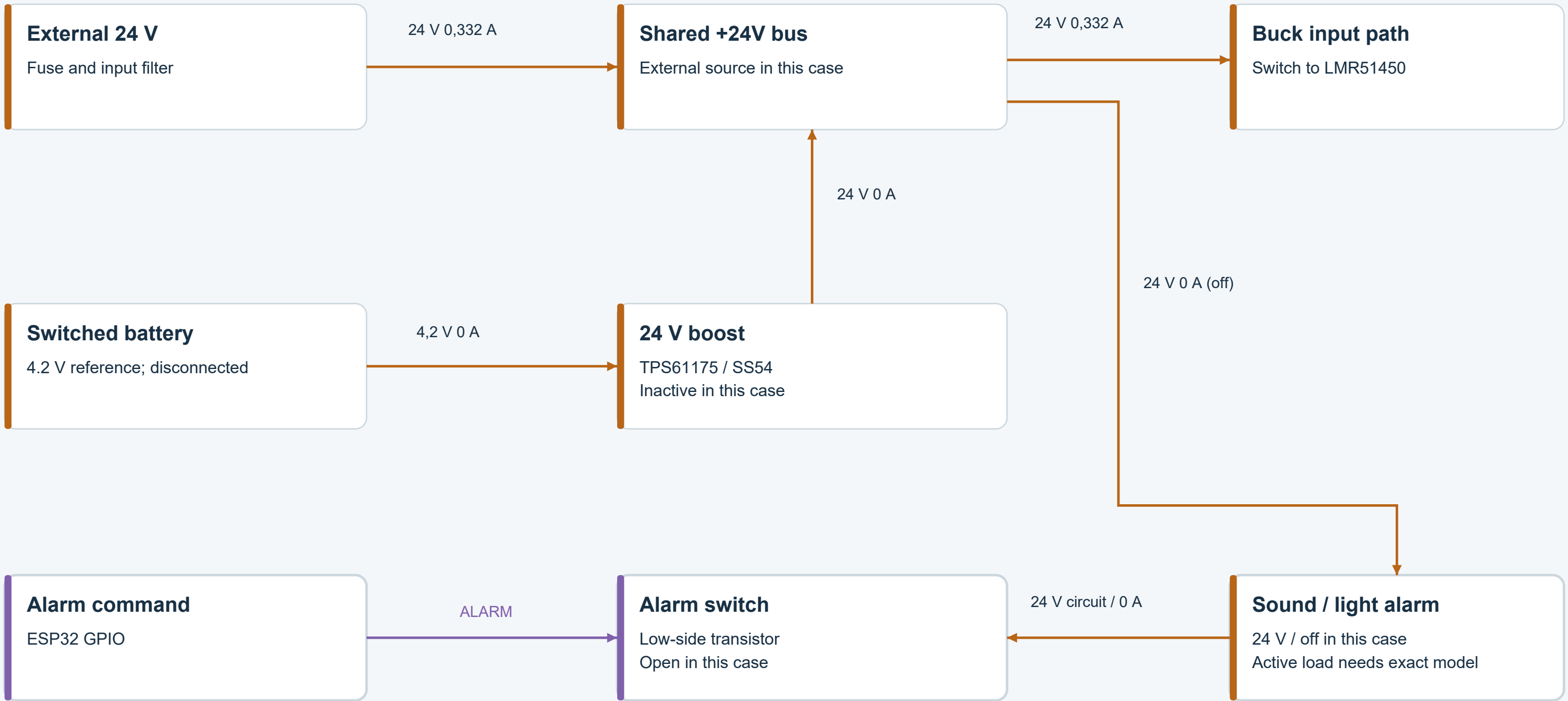
IC current references are shown inside blocks; SPI, I2C and UART lines carry data, not supply current.



VCC includes 2.409 mA pullup allowance. PCF and TCA use cross-voltage datasheet budget references; see calculation notes.

08 Shared 24 V rail and alarm

The alarm is a separate 24 V branch. Baseline values below use alarm off; battery disconnected.

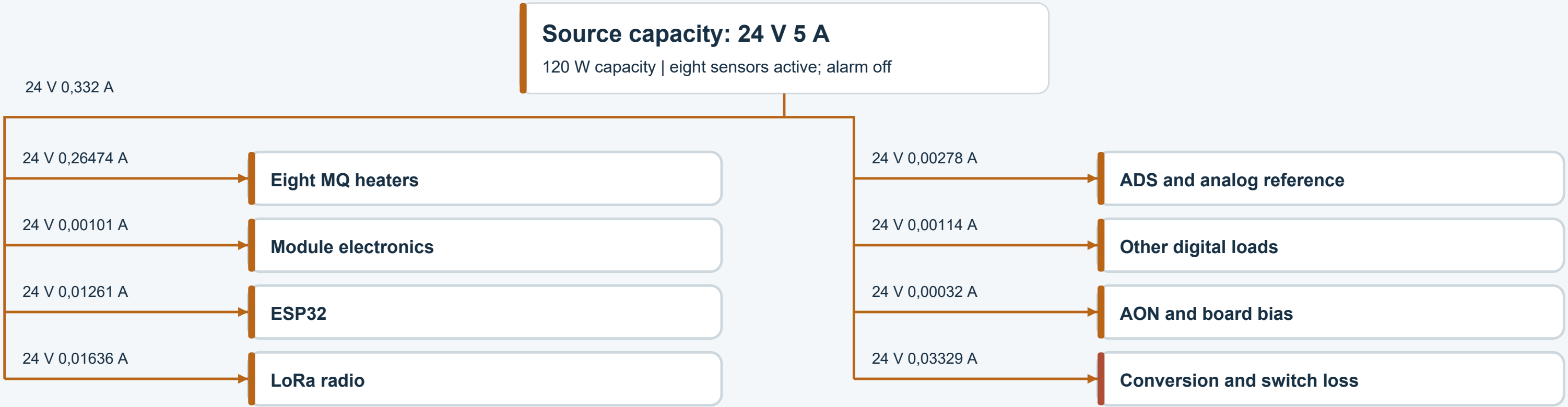


Adding the active alarm

Baseline input: about 0,332 A. Add the alarm's rated 24 V current to obtain the alarm-on estimate.
BYS / YS-01 / YS-02 does not identify one load. Alarm current and inrush are not included in the page 09 total.

09 Where the 24 V input power goes

All percentages use the full 5 A source capacity. Branch currents are power-equivalent values at 24 V.



Load / allocation	Current at 24 V	Share of 5 A	Power
Eight MQ heaters	0,26474 A	5,2948%	6,354 W
Module electronics	0,00101 A	0,0203%	0,024 W
ESP32	0,01261 A	0,2522%	0,303 W
LoRa radio	0,01636 A	0,3272%	0,393 W
ADS and analog reference	0,00278 A	0,0556%	0,067 W
Other digital loads	0,00114 A	0,0229%	0,027 W
AON and board bias	0,00032 A	0,0063%	0,008 W
Conversion and switch loss	0,03329 A	0,6657%	0,799 W
TOTAL USED	0,33225 A	6,6449%	7,97 W
REMAINING CAPACITY	4,66775 A	93,3551%	112,03 W

Reference estimate: alarm off. Active alarm, startup, unmodelled switching and input wiring/filter losses are excluded.

Remaining capacity is provisional PSU headroom, not a PCB current rating. Calculation basis: CALCULATION-NOTES.md.